

Nitrosative Stress and Retinal Neurodegeneration in Retinitis Pigmentosa Evidence from the rd10 Mouse Model

Keywords:

Retinitis pigmentosa, Nitrosative stress, Nitric oxide, NOS

Purpose:

Retinitis pigmentosa (RP) is a group of inherited retinal dystrophies characterised by progressive photoreceptor degeneration. In the rd10 mouse model, a mutation in the *PDE6 β* gene induces early retinal dysfunction followed by photoreceptor cell death. Increasing evidence indicates that oxidative and nitrosative stress contribute to retinal neurodegeneration. Nitric oxide (NO), generated by nitric oxide synthases (NOS), can exert neurotoxic effects when dysregulated. This study investigates the role of NO-mediated nitrosative stress in retinal degeneration using organotypic retinal explant cultures as an ex vivo disease model.

Approach:

A multimodal approach combining in silico transcriptomic analysis, immunohistochemistry, high-performance liquid chromatography (HPLC), and organotypic retinal explants was used. Single-cell RNA sequencing data from human and murine retinas were analysed to characterise genes involved in NO metabolism. Immunohistochemistry of rd10 and wild-type retinas at postnatal days (PN) 11, 17, and 21 assessed NOS isoforms and arginase expression. Retinal explants were treated with NOS inhibitors: L-NAME, 1400W, and 7-nitroindazole, to evaluate photoreceptor survival. Glutamate levels were quantified by HPLC to assess excitotoxicity.

Results:

NOS1 expression was mainly detected in amacrine and ganglion cells, while NOS2 showed low basal expression but was upregulated under degenerative conditions; NOS3 was restricted to vascular cells. Rd10 retinas displayed a progressive increase in NOS expression, particularly iNOS, from PN11 to PN21. Arg1 expression remained unchanged, whereas Arg2 was consistently downregulated. Elevated glutamate levels were detected in rd10 retinas. Treatment with L-NAME and 1400W significantly reduced photoreceptor cell death in a dose-dependent manner.

Conclusions:

Nitrosative stress contributes to photoreceptor degeneration in the rd10 model of RP. iNOS upregulation together with Arg2 downregulation suggests an imbalance in NO metabolism that enhances retinal neurotoxicity. Pharmacological inhibition of NO synthesis confers neuroprotection, highlighting NO-related pathways as promising therapeutic targets and supporting the utility of ex vivo retinal models for preclinical studies.